

Review Fase 1: Math Olympiad Sprint (Number Sense) - Y

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Student

Student: Yuki

Topic: Review Fase 1: Math Olympiad Sprint (Number Sense)

Status: completed

Lesson Plan

Learning Objectives

- Solve 10 elementary-level Olympiad problems focusing on number patterns and place value.
- Apply the elimination strategy to select the correct answer in multiple-choice questions.
- Justify the reasoning behind a chosen answer using mathematical terminology.
- Identify missing numbers in complex sequences up to 100.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Yuki! Ready for our Space Math Mission? First, let's fire up the engines with a quick review. Look at the screen! Task 1: What is the value of 5 in the number 528? Task 2: If I have 3 hundreds, 0 tens, and 7 ones, what number am I? Task 3: Look at these stars: 15, 16, 17... what are the next two numbers? Task 4: Which is smaller, 9 or 19?

Activities:

- *Place Value Rapid Fire: Identifying hundreds, tens, and units.*
- *Number Sequence Jump: Completing 1-20 sequences.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Great job warming up! Today, you are a Space Commander. In space, we use patterns to find hidden planets. Math Olympiad questions are like secret codes. If we learn to crack them, we can solve any mystery! Today, we will master the 'Elimination Trick' to find the right path through the stars.

Activities:

- *Theme Introduction: Space Exploration.*
- *Explanation of why patterns matter in real life (navigation/coding).*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we sprint, remember our tools. A number like 123 has 1 hundred, 2 tens, and 3 ones. And patterns always follow a rule, like +2 or -1. Tell me Yuki, if I count 2, 4, 6... what is the rule?

Activities:

- *Quick recap of 'Rule of Patterns'.*
- *Visual breakdown of a 3-digit number.*

4. Problem Introduction (10 minutes)

Teacher Script:

Now, let's look at an Olympiad Sprint question. 'Which number comes next: 10, 20, 30, ___? Options: A) 31, B) 40, C) 50.' Instead of just guessing, let's use our Elimination Shield! Can it be 31? No, because we are jumping by tens. Cross out A! Now, B or C? Let's check the rule.

Activities:

- *Modeling the Elimination Strategy.*
- *Direct instruction on identifying 'distractor' answers.*

5. Guided Practice (15 minutes)

Teacher Script:

Let's try 5 problems together. Problem 1: Look at the pattern 5, 8, 11, 14. What is the next number? Let's use our fingers to find the jump. 5... 6, 7, 8! It's a jump of 3! Now you try to find the next one. Tell me your thinking out loud!

Activities:

- *Solving 5 Olympiad-style multiple choice questions.*
- *Verbal reasoning practice (Think-Aloud).*

6. Independent Practice (15 minutes)

Teacher Script:

Commander Yuki, it's your turn to fly the ship! Solve these next 5 questions on your own. Use the drawing tool to cross out the wrong answers first. I'm here if you hit an asteroid!

Activities:

- *Solo solving of 5 number sense problems.*
- *Final 'Boss Level' question involving place value comparison (e.g., Which is larger: 3 tens or 25 ones?).*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Mission Accomplished! You were amazing at spotting patterns today. What was your favorite part? Remember, always look for the 'rule' and cross out the 'silly' answers. Don't forget to join <https://learn.algonova.id/login> to see your rewards and practice more!

Activities:

- *Recap of the Elimination Strategy.*
- *High-five and rewards.*

Homework

Medium Questions:

1. What is the missing number in the sequence: 12, 15, 18, __, 24?
2. Identify the number: 4 hundreds, 2 tens, and 9 ones.
3. Which is greater: 8 tens or 79 ones?
4. Find the rule for this pattern: 50, 45, 40, 35, 30.

Hard Questions:

1. I am a number between 20 and 30. If you add my digits, the sum is 5. What number am I?
2. Continue the pattern: 1, 2, 4, 7, 11, __.
3. How many tens are there in the number 450?
4. Which number does NOT belong: 2, 4, 6, 9, 10, 12?
5. A sequence starts at 10 and adds 5 each time. What is the 6th number in the sequence?
6. If you have two numbers, 34 and 43, which one has a 4 in the tens place?

Answer Key:

- `1. 21 (Rule: +3)`
- `2. 429`
- `3. 8 tens (80 > 79)`

- `4. Subtract 5 (-5)`
- `5. 23 (2+3=5)`
- `6. 16 (Pattern adds +1, then +2, +3, +4, +5)`
- `7. 45 tens`
- `8. 9 (The others are even numbers)`
- `9. 35 (10, 15, 20, 25, 30, 35)`
- `10. 43`
- `REVIEW: Place Value - 200` (The value of 2 in 245)
- `REVIEW: Place Value - 731` (7 hundreds, 3 tens, 1 unit)
- `REVIEW: Place Value - 50` (5 tens is 50)
- `REVIEW: Place Value - 900` (9 in hundreds place)
- `REVIEW: 1-20 - 14` (Number after 13)
- `REVIEW: 1-20 - 9` (Number before 10)
- `REVIEW: 1-20 - 12, 14, 16` (Even numbers between 10-18)
- `REVIEW: 1-20 - 15` (Midpoint of 10 and 20)

Parent Reminder

🎉 Mission Success: Yuki's Math Sprint!

Hi! Yuki did a fantastic job today as a Space Commander, tackling Math Olympiad-style number patterns and place value. We practiced the 'Elimination Strategy' to solve tricky multiple-choice questions. Please help Yuki log in to <https://learn.algonova.id/login> to complete the homework and keep the momentum going!